

A-level

ENVIRONMENTAL SCIENCE

Paper 2 - Generated practice paper Set 04

Mark scheme

120 marks: Questions 1-10 (95 marks) plus one essay from Question 11 (25 marks).

Marking guidance

- Marking points are separated by bullets; each bullet is worth 1 mark unless stated.
- Where a question asks for a fixed number of points (eg *two* reasons), credit only the first points given.
- Alternative correct wording and other relevant, accurate points are credited.
- For levels-of-response questions, first determine the level using the descriptors, then position the mark within the level using the indicative content.
- Calculations: credit correct working (method marks) even if the final answer is wrong; incorrect answers with no working score 0.

Textbook references: every question below cites the printed page numbers in R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018) from which the content is drawn, together with the AQA 7447 specification reference.

Question	Marking guidance	Marks
01.1	• Environmental DNA (eDNA) analysis	1
01.2	• Transmitters are too heavy for small birds / their batteries would not last for a long-term study • Geolocator tags are small and only record time and light levels; latitude and longitude for each day are worked out when the bird is recaptured and the data downloaded	2
01.3	• Each separate (regional) population has its own unique gene pool • DNA from seized ivory can be compared with the database to identify where the elephant came from, so poaching hotspots / trade routes can be targeted	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 404-405; AQA spec 3.1.2.3.6.	
02.1	• The new opportunity is overwhelmingly attractive (economic benefit) • Lack of knowledge / understanding of the impacts • The impacts occur elsewhere or much later, so are not evident to those who benefit	3
02.2	• Cultural belief that resources should be exploited to satisfy needs only • Problems of over-exploitation become obvious quickly as local supplies dwindle, so behaviour changes	2
02.3	• Resources are brought in from other areas • If one area is depleted, exploitation moves to another area (eg timber, fish, metals, energy) without the consumers noticing	2
02.4	• Easier access to energy has made many new activities possible • The scale of activities has greatly increased • Toxic wastes are produced that cannot be decontaminated by natural processes	3
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 370-371; AQA spec 3.6.1.	

Question	Marking guidance	Marks
03.1	- Natural cycles use a small number of abundant elements (C, H, O, N, P) built into monomers and polymers (carbohydrates, proteins); wastes of one process are raw materials for another so nothing is depleted or accumulates - Natural wastes are biodegradable / non-toxic and are broken down by decomposers into raw materials - Human systems are linear: extract - process - manufacture - use - discard; minerals are dispersed after use or mixed so recovery is difficult - Human wastes are often persistent and toxic (organochlorines, PCBs, heavy metals in electronics) and build up; fossil fuel use is inherently linear and unsustainable	4
03.2	- Metals are dispersed in small amounts within complex mixed products so they cannot easily be separated / recovered at end of life - They leach from landfill or are released by incineration; persistent, bioaccumulating	2
03.3	- Design for durability, repair and re-use (replaceable parts, no built-in batteries, spare parts and support, no planned obsolescence) - Easy identification of materials (coding) and separation of components so they can be recycled - Separation of technical materials (metals, plastics) from biological materials (which can be composted / digested) - Use of recycled and recyclable / biodegradable materials; reduced packaging and material use (Cradle to Cradle)	4
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 373-374; AQA spec 3.6.3.	
04.1	- Plants close their stomata to reduce transpiration losses - Closed stomata stop the absorption of carbon dioxide so photosynthesis and growth stop - Less water for nutrient absorption / transport and cell turgidity; severe shortage dehydrates cells and inhibits biochemical reactions	3
04.2	- Anaerobic soil favours denitrifying bacteria and inhibits nitrifying bacteria so nitrate is lost faster than it forms; higher risk of fungal disease; root respiration inhibited - Drainage ditches or pipes - Deep ploughing; avoiding compaction by machinery / livestock; adding organic matter to encourage worms	4
04.3	- Groundwater often contains dissolved salts; when the water evaporates from the soil the salts are left behind and accumulate - High soil salinity causes osmotic dehydration of roots / crop death; washing out with extra water uses more water and salinises rivers	2
04.4	Add crushed lime (calcium carbonate / hydroxide); low pH increases leaching, inhibits nutrient uptake and mobilises toxic aluminium and lead	1
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 308-310, 313; AQA spec 3.5.1.1.	
05.1	Its mass	1

Question	Marking guidance	Marks
05.2	- The 24-hour rotation period means the surface is not exposed to sunlight for long, so temperature extremes / excessive heating and cooling are reduced - The axis is tilted relative to the orbit so the amount of insolation at each latitude changes through the year, producing seasonal variations in conditions	2
05.3	- The distance from the Sun gives surface temperatures between 0 and 35 °C in most areas: too close and water would boil / evaporate, too far and it would freeze - Atmospheric pressure (produced by the mass of gases held by gravity) is high enough to stop the water boiling / evaporating away; at much lower pressure there would be no liquid water	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 5-7; AQA spec 3.1.1.1.	
06.1	• $(240\,000 - 2\,000) / 240\,000 \times 100 = 99.2\%$	1
06.2	- Whales are k-selected: slow growing, late maturing, one calf every few years, so populations recover very slowly - Technology (harpoon cannons, steam / factory ships) allowed catches far above the sustainable yield within decades - No regulation / open access to the ocean; whales were valuable (oil, spermaceti) and easily found	3
06.3	- Moratorium on commercial whaling since 1986 until populations recover - Total protection for certain species; whale sanctuaries - Limits on numbers and sizes taken; protection of suckling mothers and calves - Research into whale biology and threats (ship strikes, net entanglement); exceptions: aboriginal subsistence quotas (eg Inuit bowheads), 'scientific' permits (Japan - ruled not research in 2014), Norway / Iceland set their own quotas	4
06.4	- ITTO's 1990 target of trade only from sustainably managed forests by 2000 was not met and little progress since - Timber comes from many countries / private land with weak enforcement, high demand and alternative land uses; whaling involves few nations and a single treaty body	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 31-32; AQA spec 3.1.2.3.2.	
07.1	- Ecological footprint: the area of the Earth's surface (global hectares) needed to provide the resources used by a person / population / activity and to absorb the wastes produced - Biocapacity: the biological productivity of an area (forests, cropland, grazing, fisheries, built land) - its ability to supply resources and absorb wastes, expressed in global hectares	2
07.2	- A hectare of land / sea with the world-average biological productivity - Biocapacities of real areas vary widely so a standard unit allows footprints and biocapacities of different places to be compared	2
07.3	• $8.1 / 1.7 = 4.8$ (accept 4.7-4.8)	1

Question	Marking guidance	Marks
07.4	- Higher HDI is associated with a larger footprint (USA 8.1 vs Bangladesh 0.8), though the UK achieves the same HDI as the USA with about half the footprint - Global footprint (2.8) exceeds global biocapacity (1.7) so humanity uses resources faster than they regenerate and CO₂ accumulates (overshoot) - Earth Overshoot Day is the date each year when a year's worth of resources / waste absorption has been used up; it falls earlier each year (8 August in 2016)	3
07.5	<i>Any 2 from:</i> - Renewable energy and energy conservation (carbon footprint is the largest component) - Reduced meat consumption / food waste; local seasonal food - Circular economy: repair, re-use, recycling; longer product lives - Public transport, efficient buildings	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 379-380; AQA spec 3.6.4.2.	
08.1	- South America: high loss (5.1-5.5 Mha yr⁻¹) falling by about half by 2020 - Africa: loss steady at about 3.4 then rising to 3.9 by 2020 - now the highest - Europe: negative values - forest area increasing (afforestation exceeds loss), though the gain is shrinking	3
08.2	- South America: clearance for commercial cash crops (sugar cane, soya) and cattle ranching for export; HEP reservoirs; roads improving access; mining - Africa: growing rural populations clearing land for subsistence farming; fuelwood and charcoal collection above regrowth (charcoal wastes much of the wood's energy); fodder cutting in dry areas - Both: hardwood timber (teak, mahogany) from clear-felled mixed forest often not replanted; urban expansion	4
08.3	- They create barriers fragmenting wildlife populations - Improved access makes it easier to remove timber and farm produce, so clearance spreads along them	2
08.4	Conifers grow quickly so plantations are replanted after harvest and yield a saleable crop within decades; slow-growing hardwoods (teak, mahogany) take too long for commercial plantations so they are taken from natural forest that is clear-felled and not replanted	1
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 367-368; AQA spec 3.5.3.3.	
09.1	- Overall decline from 62% to 22% - Sharp falls in 1998 (to 35%) and 2016 (to 18%) coinciding with bleaching events, with partial recovery after each	2
09.2	- The polyps expel their symbiotic zooxanthellae so the coral loses colour; bleached corals are alive but grow very slowly and may die if algae do not return - Raised water temperature (climate change / El Niño) - Pollution: sewage, pesticides, cyanide, sunscreen; low oxygen from zooplankton blooms after overfishing - High light / solar activity; ocean acidification hinders skeleton formation	4

Question	Marking guidance	Marks
09.3	• Coastal construction and dredging increase sedimentation / turbidity covering and killing polyps • Sewage and fertiliser nutrients cause algal growth that shades corals and prevents egg release • Physical damage from anchors, divers, souvenir collection; loss of mangroves / seagrass nurseries that trap sediment	3
09.4	• Protected breeding populations grow larger and produce surplus young / adults that spill over into fished areas	1
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 60-63; AQA spec 3.1.2.3.4.	
10.1	• About 10% of land is already cultivated and 25% grazed • Most remaining land is too dry, wet, cold, hot or steep, is covered in ice, or has no soil	2
10.2	• Land becomes too expensive for local people to rent / is used to grow export crops for more affluent societies • So people with less money lose access to the land and food they used to produce	2
10.3	• It uses natural processes to supply nutrients (legumes, recycled organic matter, soil biota), control pests and provide ecological services • So that pesticides and artificial fertilisers become unnecessary rather than just being banned	2

Question	Marking guidance	Marks
10.4	Levels of response Level 3 (7-9 marks): Answer is comprehensive, relevant and logically structured. Shows detailed knowledge and understanding of the relevant scientific ideas and applies them accurately to the context. Evaluation/analysis is well supported by evidence and a clear judgement or conclusion is reached. Level 2 (4-6 marks): Answer is mostly relevant and shows some structure. Knowledge and understanding of relevant ideas is sound but application to the context or evaluation may be incomplete or lack detail. A judgement may be made but is not fully supported. Level 1 (1-3 marks): Answer is basic, with limited structure. Some relevant knowledge is shown but there are omissions or errors. Little application to the context and little or no evaluation. 0 marks: No relevant content. Indicative content (credit other relevant, accurate points) • Unsustainable features: reliance on chemical pesticides (resistance, non-target impacts, bans), non-renewable rock phosphate, fossil-fuel-intensive nitrate manufacture, over-exploitation of rivers / groundwater, salinisation, loss of soil organic matter, greenhouse gases from fuel, rice and livestock, loss of species providing ecosystem services • Pest control: cultural methods (weeding, mulching, crop rotation, barrier crops, culling), biological control, predator habitats, polyculture / companion crops, integrated control, reduced antibiotic use • Nutrients: recycling organic matter, crop rotation, cultivation of legumes, conservation of soil biota, reduced artificial fertiliser especially nitrate, low tillage • Water: low water-use crops, reservoirs and aquifer recharge, drip irrigation rather than sprays • Soil and carbon: low-tillage farming to maintain organic matter, reduced machinery, renewable energy, drought-tolerant rice varieties so fields are drained earlier, high-carbohydrate ground feed for cattle reduces methane • Biodiversity: conserve habitats of crop wild relatives, seed banks, retain hedgerows, ditches, ponds, woodland; agri-environment schemes • Evaluation: growing population (perhaps 12 billion) needs more food; lower trophic level diets; obstacles - cost, need for support, gaining acceptance of new techniques; learning from past land degradation (fertile crescent, US mid-west) • Judgement: no single strategy sufficient; combination of ecological principles and new technology; justified conclusion	9
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 335-337; AQA spec 3.5.1.5.	

Question	Marking guidance	Marks
11.1	Explain how the physical features of the Earth and the activities of living organisms have combined to create and maintain conditions suitable for life. Levels of response Level 5 (21-25 marks): Comprehensive and detailed knowledge and understanding drawn from across the specification, applied accurately to the question. Argument is coherent, well-structured and supported by a wide range of relevant, well-chosen examples and quantitative detail where appropriate. Interconnections between topics are made explicitly and a clear, justified conclusion is reached. Level 4 (16-20 marks): Detailed knowledge and understanding with good application to the question. Argument is logically organised with relevant examples; some links between different areas of the specification are made. Conclusions are drawn but may be less fully justified. Level 3 (11-15 marks): Sound knowledge and understanding, mostly applied to the question. Some structure and relevant examples, but coverage may be uneven or descriptive rather than analytical. Limited linking of topics. Level 2 (6-10 marks): Basic knowledge and understanding with partial relevance. Answer is largely descriptive with few examples and limited structure; some errors or omissions. Level 1 (1-5 marks): Limited, fragmented knowledge with little relevance to the question. Little structure; examples absent or inappropriate. 0 marks: No relevant content. Indicative content (students are not expected to cover all of these; credit other relevant material) • Physical features: mass / gravity retaining an atmosphere and giving pressure for liquid water; distance from the Sun and the Goldilocks zone; speed of rotation reducing temperature extremes; tilt producing seasons; magnetic field deflecting the solar wind; the Moon stabilising the orbit and producing tides • Importance of liquid water: solvent, transport, temperature control, anomalous expansion, high specific heat capacity, habitats, absorption of UV before the ozone layer • Development of life: Archaea at vents; photosynthesis; oxygen reacting with iron then accumulating; ozone layer allowing colonisation of land; aerobic organisms • Life changing conditions: carbon sequestration offsetting solar brightening, biogeochemical cycles, transpiration and rainfall, regulation of atmospheric gases • Methods of studying past conditions: proxy data, isotope ratios, ice cores, Mg:Ca ratios; understanding of continental drift and ocean currents • Links to current threats (climate change, ozone depletion) and a justified conclusion about the interdependence of physical and biological factors Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 5-9; AQA spec 3.1.1.	25

Question	Marking guidance	Marks
11.2	Discuss the extent to which the resources and services provided by other species justify the conservation of biodiversity. Levels of response Level 5 (21-25 marks): Comprehensive and detailed knowledge and understanding drawn from across the specification, applied accurately to the question. Argument is coherent, well-structured and supported by a wide range of relevant, well-chosen examples and quantitative detail where appropriate. Interconnections between topics are made explicitly and a clear, justified conclusion is reached. Level 4 (16-20 marks): Detailed knowledge and understanding with good application to the question. Argument is logically organised with relevant examples; some links between different areas of the specification are made. Conclusions are drawn but may be less fully justified. Level 3 (11-15 marks): Sound knowledge and understanding, mostly applied to the question. Some structure and relevant examples, but coverage may be uneven or descriptive rather than analytical. Limited linking of topics. Level 2 (6-10 marks): Basic knowledge and understanding with partial relevance. Answer is largely descriptive with few examples and limited structure; some errors or omissions. Level 1 (1-5 marks): Limited, fragmented knowledge with little relevance to the question. Little structure; examples absent or inappropriate. 0 marks: No relevant content. Indicative content (students are not expected to cover all of these; credit other relevant material) • Resources: wood, fibres, oils, fuels, food; new food species (morama bean, yeheb, perennial maize, Kernza, bison, eland) and their adaptations • Biomimetics: wing-tip feathers, whale tubercles, shark skin, termite mounds, bird bones, gecko adhesion, burrs, lotus, spider silk • Medicines: poppies, cinchona, willow, yew (taxol), Mexican yam, sponges (AZT), sharks; few species yet studied; physiological research (marsupials, hippo, sponges) • Genetic resources: crop wild relatives, Vavilov centres, gene pools and inbreeding • Ecosystem services: atmospheric regulation, hydrological cycle, biogeochemical cycles, soil maintenance; interspecies relationships - food, pollination (Darwin's orchid), seed dispersal, habitat provision • Evaluation: economic vs intrinsic / moral arguments; services unnoticed because of dynamic equilibrium; unknown future value; justified conclusion Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 10-19; AQA spec 3.1.2.1.	25